

Lecture 10 Erdős - Ko-Rado & Fractional Helly

§1 Erdős - Ko-Rado.

Setup: $\mathcal{F} = \{F_1, F_2, \dots, F_s\} \subseteq 2^{[n]}$ is
intersecting if sets in $\mathcal{F}$ are pairwise intersecting.
i.e. $\forall F_i, F_j \in \mathcal{F}, F_i \cap F_j \neq \emptyset$.

Consider k -unif intersecting family $\mathcal{F} \subseteq \binom{[n]}{k}$

Question How large can $\mathcal{F}$ be? all k -sets in $[n]$

- May assume $n \geq 2k$

Construction (Star) k -unif family $\mathcal{F}$ s.t. $\exists x \in [n]$
 $\forall F \in \mathcal{F}, x \in F$

size of a star

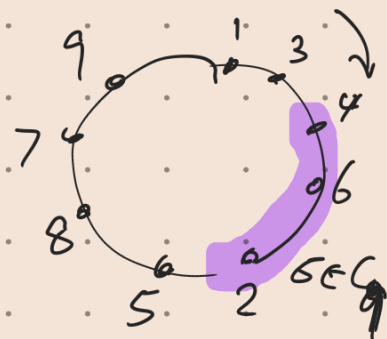
$$= \binom{n-1}{k-1}$$



Thm (Erdős - Ko-Rado 603) Let $n \geq 2k$. If $\mathcal{F} \subseteq \binom{[n]}{k}$
is intersecting, then $|\mathcal{F}| \leq \binom{n-1}{k-1}$

Pf (Katona's pf) Idea: double count the pairs
 (σ, F) , $\sigma \in C_n$, $F \in \mathcal{F}$ s.t.

F appears on σ as an interval
(i.e. elements of F appear consecutively on σ)



$k=3, F=\{2,4,6\}$

Shall see that

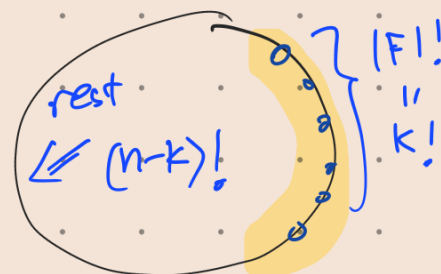
$$|C_n| \cdot k$$

$$|F| \cdot k! \cdot (n-k)! = \# \text{ pairs} = P \leq (n-1)! \cdot k \dots (*) (\sigma, F)$$

$$\Rightarrow |F| \leq \binom{n-1}{k-1}$$

• " $=$ " in $(*)$: Count (σ, F) from F eg pt of view.

$$P = |F| \cdot k! \cdot (n-k)! \quad \begin{array}{l} \nwarrow \text{ways to arrange } F \text{ eg} \\ \nearrow \text{arrange the rest} \end{array}$$



• " $\leq$ " in $(*)$

Enough to show: Fix a $\sigma \in C_n$, there are at most k members of F contained in σ as intervals.

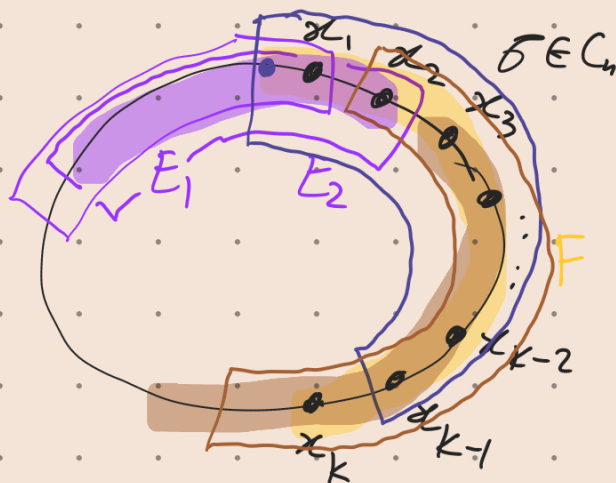
$$(\sigma, F) \leq k \text{ choices}$$

• Further fix a set $F \in \mathcal{F}$ contained as an interval in σ , say $(x_1, x_2, \dots, x_k)$

• $\forall 1 \leq i \leq k-1,$

$E_i = \text{length-}k \text{ interval on } \sigma \text{ ending at } x_i$

• $\forall 2 \leq i \leq k, S_i = \text{length-}k \text{ interval starting at } x_i$



• as $\mathcal{F}$ is intersecting & $F \in \mathcal{F}$, then any interval in σ that belongs to $\mathcal{F}$ must be one of the $\{S_i\}_{2 \leq i \leq k}$ & $\{E_i\}_{1 \leq i \leq k-1}$.

• Note that $\forall 1 \leq i \leq k-1, E_i \cap S_{i+1} = \emptyset$
 $\Rightarrow \mathcal{F}$ contains at most one of them.

$\Rightarrow$ # intervals in σ belonging to $\mathcal{F} \leq 1 + k-1 = k$. $\square$

Exer $n \geq 2k+1$, then stars are unique extremal examples.

Exer if $n = 2k$, give a non-star extremal example.

• Non-uniform, $|\text{star}| = 2^{n-1}$



Exer if $\mathcal{F} \in 2^{[n]}$ intersecting

$$\Rightarrow |\mathcal{F}| \leq 2^{n-1}$$

Exer Give a non-star extremal example.

Recall

(set-pairs lineq) $A_1, \dots, A_s, B_1, \dots, B_s$ s.t.

$$|A_i| \leq a, |B_i| \leq b$$

$$\forall i, A_i \cap B_i = \emptyset$$

$$\forall i \neq j, A_i \cap B_j \neq \emptyset$$

(skewed) $\forall i < j$

$$\Rightarrow s \leq \binom{a+b}{a}$$

Application to counting k -unif maximal intersecting families
(under set inclusion)

• $|\text{Star}| = \binom{n-1}{k-1}$ & any subfam of a star still intersecting
 $\Rightarrow \# \text{ } k\text{-unif intersecting families} \geq 2^{\binom{n-1}{k-1}}$

• B-D-D-L-Sh 2015: if $n \geq 2k+1$, then
 $\# \text{ } k\text{-unif inters. fam.} = 2^{(1+o(1)) \binom{n-1}{k-1}}$

Exer if $n = 2k$, there are exponentially more:
 $\geq 3^{\binom{n-1}{k-1}}$ many.

On contrast, far fewer maximal intersecting families.

Thm $\# \text{ maximal } k\text{-unif intersecting fam. on } [n]$
 $\leq \sum_{i=0}^{\binom{2k}{k}} \binom{\binom{n}{k}}{i} \leq \binom{n}{k}^{\binom{2k}{k}}$

Idea: $\begin{array}{ccc} \text{maximal int. fam} & \xleftrightarrow{1\text{-to-1}} & \text{generating set } \mathcal{G} \subseteq 2^{[n]} \\ \# & \xleftarrow{\text{same}} & \# \end{array}$
 $\# \mathcal{G} \leq \binom{2k}{k}$

Def: Given a $\mathcal{F} \subseteq \binom{[n]}{k}$, write

$\mathcal{I}(\mathcal{F}) = \left\{ G \in \binom{[n]}{k} : \forall F \in \mathcal{F}, G \cap F \neq \emptyset \right\}$

i.e. $\mathcal{I}(\mathcal{F})$ consists of all k -sets that intersect all members of $\mathcal{F}$.

Obs 1) $\mathcal{F}$ is intersecting $\Leftrightarrow \mathcal{F} \subseteq \mathcal{I}(\mathcal{F})$

2) $\mathcal{F}$ is maximal intersecting $\Leftrightarrow \mathcal{F} = \mathcal{I}(\mathcal{F})$

Given a maximal intersecting fam $\mathcal{F}$, we call

$G \subseteq \mathcal{F}$ a generating set if $\mathcal{F} = \mathcal{I}(G)$

Pf: • Map each $\mathcal{F}$ maximal int. fam $\rightarrow \mathcal{F}_0 \subseteq \mathcal{F}$
i.e. $\mathcal{I}(\mathcal{F}_0) = \mathcal{F}$ minimal generating set.

& $\forall \mathcal{F}'_0 \subsetneq \mathcal{F}_0, \mathcal{I}(\mathcal{F}'_0) \subsetneq \mathcal{F}$

this is an injective map.

• Left to show that $\forall \mathcal{F}$ max. int. fam.

its minimal gen. set $|\mathcal{F}_0| \leq \binom{2k}{k}$

$$\Rightarrow \# \mathcal{F} = \# \mathcal{F}_0 \leq \# \text{set systems of size} \leq \binom{2k}{k} \\ = \sum_{i=0}^{\binom{2k}{k}} \binom{\binom{2k}{k}}{i}$$

• Let $\mathcal{F}_0 = \{F_1, F_2, \dots, F_s\} \subseteq \mathcal{F}$ min. gen. set.

By minimality, $\forall F_i \in \mathcal{F}_0$

$$\Rightarrow \mathcal{F} \subsetneq \mathcal{I}(\mathcal{F}_0 \setminus \{F_i\})$$

i.e. $\exists$ some $G_i \in \mathcal{I}(\mathcal{F}_0 \setminus \{F_i\}) \setminus \mathcal{F}$ s.t.

$$G_i \cap F_i = \emptyset$$

$$\& \quad G_i \cap F_j \neq \emptyset \quad \forall j \neq i$$

$$\{F_i\}_{i \in [s]} \& \{G_i\}_{i \in [s]}$$

$$\text{Set-pairs inequality} \Rightarrow s \leq \binom{k+k}{k} = \binom{2k}{k}$$

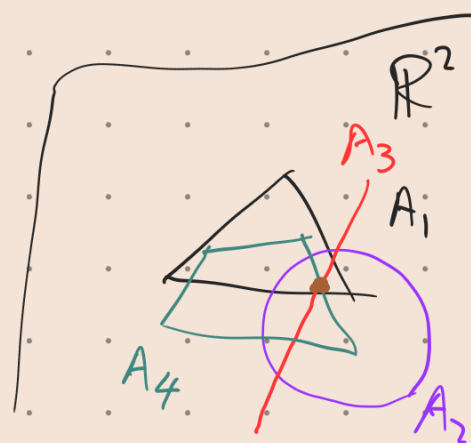
Fractional Helly

A collection of convex sets

Def : $\mathcal{A} = \{A_1, A_2, \dots, A_k\}$ in $\mathbb{R}^d$ is

pierced by a pt. if $\bigcap_{i \in [k]} A_i \neq \emptyset$

ii $\bigcap \mathcal{A}$



Fractional
(Helly's thm) local - global

$\forall \alpha \in (0,1)$

Let $\mathcal{F}$ be a collection of convex sets $\bigcap \mathcal{A} \neq \emptyset$

in $\mathbb{R}^d$. If $\geq \alpha$ fractional of every $(d+1)$ -tuple $\mathcal{A} \in \binom{\mathcal{F}}{d+1}$ is pierced

by some pt, then $\bigcap \mathcal{F} \neq \emptyset \Rightarrow \exists \mathcal{F}' \subseteq \mathcal{F}$ s.t.

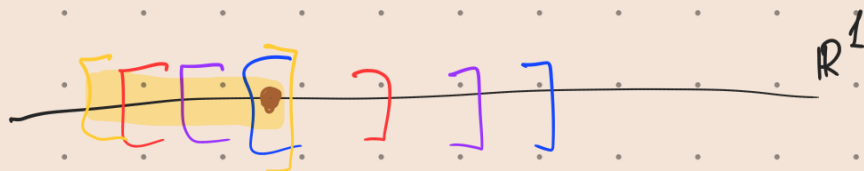
($\bigcap \mathcal{A} \neq \emptyset$)

- $\bigcap \mathcal{F}' \neq \emptyset$
- $|\mathcal{F}'| \geq \beta \cdot |\mathcal{F}|$, where

Ex $d=1$

conv sets = interval

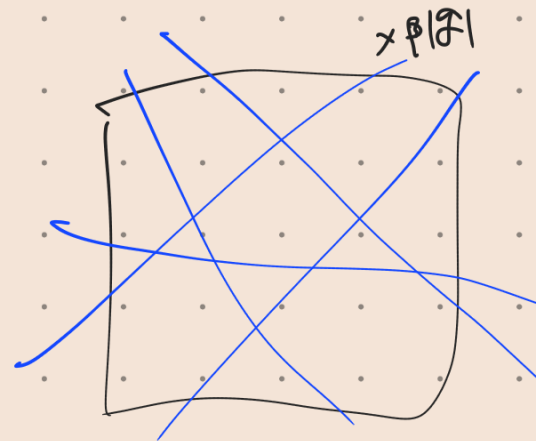
$$\beta \geq (1 - (1-\alpha)^{\frac{1}{d+1}})$$



Rmk: The bd $\beta \geq (1 - (1-\alpha)^{\frac{1}{d+1}})$ is tight.

• Consider $d=2$, $\beta = (1 - (1-\alpha)^{\frac{1}{3}})$

$\mathcal{F} = \{ \beta |\mathcal{F}| \text{ many } \mathbb{R}^2, \\ (1-\beta)|\mathcal{F}| \text{ take lines} \\ \text{in general position} \}$



• # triples w./ non-empty intersection

$$(\heartsuit) = \binom{|\mathcal{F}|}{3} - \binom{(1-\beta)|\mathcal{F}|}{3}$$

• $\mathcal{F}' \subseteq \mathcal{F}$ be those $\beta |\mathcal{F}| \mathbb{R}^2$

$$\frac{(\heartsuit)}{\binom{|\mathcal{F}|}{3}} \approx \alpha$$

(Colorful Helly) Let $\mathcal{F}_1, \dots, \mathcal{F}_{d+1}$ be collections of conv sets in $\mathbb{R}^d$. If every rainbow $(d+1)$ -tuple $\mathcal{A}$ (i.e. one set from each $\mathcal{F}_i, i \in [d+1]$) satisfy $\cap \mathcal{A} \neq \emptyset$

$\Rightarrow \exists i \in [d+1]$ s.t. $\cap \mathcal{F}_i \neq \emptyset$.